higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 4

(6050364)

18 November 2019 (X-Paper)
09:00–12:00

Scientific calculators may be used.

This question paper consists of 5 pages.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use 3,14 for the value of π
5. Start each section on a NEW page.
6. Use only a pen with BLUE or BLACK ink.
7. Write neatly and legibly.

Copyright reserved

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 The product that remains after evaporation has taken place is known as …

A a solute
B a solvent
C a concentrate
D viscous

1.2 The formation of water droplets on the outside surface of a glass filled with cold water is a typical example of …

A evaporation
B condensation
C sublimation
D decantation

1.3 The technique used in packed columns to "scrub" the gas is by …

A dividing a liquid into a number of thin films that flow over a gaseous medium.
B spraying the liquid over the gas.
C breaking off the gas into tiny bubbles.
D gas enters the column at the circumference of the column.

1.4 With the principle of material flow in a simple closed process system, mass per second is commonly known as mass flow per second and it is denoted by the symbol …

A m/s
B m/t
C Qm
D pV

1.5 What happens to the energy of a system when the system is in a steady state?

A The energy increases
B The energy decreases
C The energy becomes directly proportional to heat
D The energy of the system doesn't change

(5 × 2) **[10]**

QUESTION 2

2.1 Metals that are used in the construction of process equipment consist of atomic particles.

2.1.1 What are these atomic particles known as?

2.1.2 Which electrical polarity do these particles as mentioned in QUESTION 2.1.1 have?

(2 × 2) (4)

2.2 When is the casting process being used in the manufacturing industry? (2)

2.3 Why is galvanised steel coated with layers of zinc? (2)

2.4 Name TWO types of liquid adhesives that can be used to combine materials together. (2)

2.5 The flashing process in a flash evaporator can be broken down into THREE separate phases. Name these THREE phases. (3)

2.6 Draw a schematic block diagram of a forced circulation evaporator. (6)

2.7 Explain the operation of a plate condenser. (11)

2.8 Tabulate the difference between a *venturi scrubber* and a *cyclone scrubber*. (4)

[34]

QUESTION 3

3.1 Define the following terms:

3.1.1 Heat loss

3.1.2 Boiling point

(2 × 2) (4)

3.2 Name THREE methods that can be used to transfer heat from one substance to another substance. (3)

3.3 Write down the formulas that can be used to calculate each of the following types of energy within a system:

3.3.1 Kinetic energy

3.3.2 Potential energy

(2 × 3) (6)

3.4 Explain when does heat in a system has a negative value and when does heat in a system has a positive value. (4)

3.5 Given the heat capacity of water as 0,39 kJ/K, calculate the amount of energy required to raise the temperature of an object by 38,6 °C. (6)

3.6 Calculate the volume concentration if 250 litres of $CaCO_3$ are dissolved in 10 kg of water. (5)

[28]

QUESTION 4

4.1 State the law of conservation of mass. Also, write down an equation to express this law. (5)

4.2 Define the following terms:

4.2.1 Closed system with reference to conservation of mass

4.2.2 Volume with reference to conservation of mass

4.2.3 Molar concentration of a substance

4.2.4 Chemical reaction

(4 × 2) (8)

4.3 Define the term *atomic weight* of an element. (2)

4.4 Write brief notes on moles as they relate to the Avogadro number. (4)

4.5 The weight of a cuboidal stone is 11,4 g. Its height is 3,3 cm, its length is 4,2 cm and its breadth is 1,7 cm.

Calculate the specific weight of this stone. (9)

[28]

TOTAL: 100

Copyright reserved